SIMATS ENGINEERING

ASSESSMENT TOOL 2: Schema Analysis and Viva Voce

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO2: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)

Assessment Tool Weightage: 35%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 20**
Question Count: 2

Code of Conduct

I Naresh M (1921424381) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Naresh M

Database Design and Connectivity Rubric

Requirement Analysis (4 Marks)	ER Diagram Design (4 Marks)	Table Design & Normalization (4 Marks)	Database Connectivity using Python (4 Marks)	CRUD Operations (4 Marks)	Total (20 Marks)

Total Marks Awarded:

Questions:

Set B

Question 7: Library Management Database (10 Marks)

A library maintains the following database schema:

- Member (Member_ID, Member_Name, Phone)
- Book (Book_ID, Book_Title, Author)
- Issue (Issue_ID, Member_ID, Book_ID, Issue_Date)

Analyze the schema and write a Python program to retrieve the member name, book title, and issue date for all books issued after 1 January 2026.

Question 8: Hotel Reservation Database (10 Marks)

A hotel maintains the following database schema:

- Customer (Customer_ID, Customer_Name, City)
- Room (Room_ID, Room_Type, Rent)
- Reservation (Reservation_ID, Customer_ID, Room_ID, Check_In_Date, Check_Out_Date)

Analyze the schema and write a Python program to retrieve the customer name, room type, and check-in date for customers who have reserved Deluxe rooms.

Signature of the Course Faculty

Question 7: Library Management Database (10 Marks)

Aim:

To create a Library Management Database using SQLite and retrieve the **member name, book title, and issue date** for all books issued **after 1 January 2026**.

Procedure:

1. Import the sqlite3 module.
2. Create a database connection.
3. Create the tables Member, Book, and Issue.
4. Insert sample records.
5. Execute an SQL query using JOIN.
6. Display the member name, book title, and issue date.
7. Close the database connection.

Code:

```
import sqlite3

conn = sqlite3.connect("library.db")

cursor = conn.cursor()

cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Member(
    Member_ID INTEGER PRIMARY KEY,
    Member_Name TEXT,
    Phone TEXT
)
""")
cursor.execute("""
CREATE TABLE IF NOT EXISTS Book(
    Book_ID INTEGER PRIMARY KEY,
    Book_Title TEXT,
    Author TEXT
)
""")
cursor.execute("""
CREATE TABLE IF NOT EXISTS Issue(
    Issue_ID INTEGER PRIMARY KEY,
    Member_ID INTEGER,
    Book_ID INTEGER,
    Issue_Date TEXT,
    FOREIGN KEY(Member_ID) REFERENCES Member(Member_ID),
    FOREIGN KEY(Book_ID) REFERENCES Book(Book_ID)
)
""")
cursor.execute("INSERT INTO Member VALUES (1,'Rahul','9876543210')")
cursor.execute("INSERT INTO Member VALUES (2,'Priya','9123456780')")
cursor.execute("INSERT INTO Book VALUES (101,'Python Programming','John')")
cursor.execute("INSERT INTO Book VALUES (102,'Database Systems','Thomas')")
cursor.execute("INSERT INTO Issue VALUES (1,1,101,'2026-02-15')")
cursor.execute("INSERT INTO Issue VALUES (2,2,102,'2025-12-20')")
```

```
conn.commit()

cursor.execute("""

SELECT Member.Member_Name,

       Book.Book_Title,

       Issue.Issue_Date

FROM Issue

JOIN Member ON Issue.Member_ID = Member.Member_ID

JOIN Book ON Issue.Book_ID = Book.Book_ID

WHERE Issue.Issue_Date > '2026-01-01'

""")

records = cursor.fetchall()

print("Books Issued After 1 January 2026")

for row in records:

    print(row)

conn.close()
```

Output:

```
========= RESTART: C:/Users/nares/Videos/dsa0503/experiments/ass 2.1.py =======
Books Issued After 1 January 2026
('Rahul', 'Python Programming', '2026-02-15')
```

Question 8: Hotel Reservation Database (10 Marks)

Aim

To create a Hotel Reservation Database using SQLite and retrieve the **customer name, room type, and check-in date** for customers who reserved **Deluxe rooms**.

Procedure

1. Import the sqlite3 module.
2. Create a database connection.
3. Create Customer, Room, and Reservation tables.
4. Insert sample records.

5. Execute an SQL JOIN query.
6. Display the customer name, room type, and check-in date.
7. Close the database connection.

Code:

```
import sqlite3

conn = sqlite3.connect("hotel.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Customer(

    Customer_ID INTEGER PRIMARY KEY,

    Customer_Name TEXT,

    City TEXT

)

""")

cursor.execute("""

CREATE TABLE IF NOT EXISTS Room(

    Room_ID INTEGER PRIMARY KEY,

    Room_Type TEXT,

    Rent REAL

)

""")

cursor.execute("""

CREATE TABLE IF NOT EXISTS Reservation(

    Reservation_ID INTEGER PRIMARY KEY,

    Customer_ID INTEGER,

    Room_ID INTEGER,

    Check_In_Date TEXT,

    Check_Out_Date TEXT,

    FOREIGN KEY(Customer_ID) REFERENCES Customer(Customer_ID),
```

```
    FOREIGN KEY(Room_ID) REFERENCES Room(Room_ID)
)
""")
cursor.execute("INSERT INTO Customer VALUES (1,'Arun','Chennai')")
cursor.execute("INSERT INTO Customer VALUES (2,'Meena','Madurai')")
cursor.execute("INSERT INTO Room VALUES (101,'Deluxe',3500)")
cursor.execute("INSERT INTO Room VALUES (102,'Standard',2000)")
cursor.execute("INSERT INTO Reservation VALUES (1,1,101,'2026-03-10','2026-03-12')")
cursor.execute("INSERT INTO Reservation VALUES (2,2,102,'2026-03-15','2026-03-18')")
conn.commit()
cursor.execute("""
SELECT Customer.Customer_Name,
       Room.Room_Type,
       Reservation.Check_In_Date
FROM Reservation
JOIN Customer ON Reservation.Customer_ID = Customer.Customer_ID
JOIN Room ON Reservation.Room_ID = Room.Room_ID
WHERE Room.Room_Type = 'Deluxe'
""")
records = cursor.fetchall()
print("Customers Who Reserved Deluxe Rooms")
for row in records:
    print(row)
conn.close()
```

Output:

```
========= RESTART: C:/Users/nares/Videos/dsa0503/experiments/ass 2.2.py ========
Customers Who Reserved Deluxe Rooms
('Arun', 'Deluxe', '2026-03-10')
```